Citation

Margaret Douglas, Marshall Dozier, Emilie McSwiggan, Thulani Ashcroft, Ziwen Sun, Xiaomeng Zhang, Michael Tornow, Kayla Ostrishko, Prerna Krishan. Systematic Review of the association between transmission of respiratory viruses, including Covid19, and neighbourhood design. PROSPERO 2020 CRD42020212949 Available from: https://www.crd.york.ac.uk/prospero/display_record.php?ID=CRD42020212949

Review question

How is transmission of respiratory viruses, including Covid19, affected by features of neighbourhood design?

Searches

MEDLINE, Embase, Web of Science, Scopus

English language

Chinese databases to be piloted

Types of study to be included

Include: Studies providing empirical data: longitudinal, observational, cohort, ecological

Exclude: Opinion pieces, modelling studies regarding prediction of infection without new empirical data, qualitative studies and reviews (will retain the latter two types of study for separate analysis and background)

Condition or domain being studied

Include: Transmission of respiratory infectious diseases

Exclude: Studies of non-respiratory or non-infectious diseases

Participants/population

General population

Intervention(s), exposure(s)

Include:

Studies comparing transmission by aspects of housing/neighbourhood design:
Dwelling or population density at neighbourhood/city level

Housing type

Street connectivity or walkability

Land use mix

Features of ‘smart urbanism’ or ‘15 -20 minute neighbourhoods’

Exclude:

Overcrowding in home or household composition (bedroom number, number of adults)

Studies concerned with building materials, air and water quality

Design of healthcare, education or transport systems only

Studies assessing social mobility or effects of social distancing

Comparator(s)/control
not applicable

Context
Include: Studies in high/middle income countries

Exclude: Studies based in low income countries

Main outcome(s)
Differences in incidence/prevalence/morbidity/mortality related to transmission of respiratory infectious diseases

Measures of effect

incidence and mortality rates

Additional outcome(s)
None

Measures of effect

None

Data extraction (selection and coding)

Study selection

Each title and abstract will be screened by two reviewers. If either reviewer wishes to retain the article, it will be retained for full text screen.

Two reviewers will then screen full texts. Disagreements will be resolved through discussion.

We will also manually check the references of included studies and perform forward citation tracking to identify any further studies eligible for inclusion.

Data extraction

We will extract the following data from the included studies. For each paper, a single reviewer will extract data into a piloted form and a second reviewer will check completeness and accuracy of extracted data.

Study details:

Author

Title

Date

Study design

Country

Neighbourhood details:

Location/ setting

Population (subgroup information – income level, ethnicity, gender, SES)

Time period studied

Measures of density of population/neighbourhood

Measures of neighbourhood design

Measures of housing design

Respiratory infection details:

Type of infection

Measures of infection outcome

Main findings

Number of people: total, in outcome group and in exposure group

Analysis metric: odds ratio, percentages, morbidity, mortality,

Effect size and P value

Confounders

Risk of bias (quality) assessment

We will use the relevant CASP tool for the study type if available. For cross sectional studies we will use Joanna Briggs Institute tool.

We will use the GRADE system to rate evidence quality.

Strategy for data synthesis
We will mainly present a narrative summary of findings, with summary tables and figures. If three or more articles investigating the association between the same exposure and the same outcome are detected, we will perform a meta-analysis to analyse the summary effect size and its confidence interval by using DerSimonian and Laird (DL) estimator. The meta-analysis P value threshold was set at 0.05. We will quantify the heterogeneity of each meta?analysis by calculating the I^2 value.

Analysis of subgroups or subsets
None

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Organisational affiliation of the review
University of Edinburgh

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Type and method of review
Narrative synthesis, Systematic review

Anticipated or actual start date
13 October 2020

Anticipated completion date
30 November 2020

Funding sources/sponsors
None

Conflicts of interest

Language
Chinese-Simplified, English

Country
Scotland

Stage of review
Review Completed published

Subject index terms status
Subject indexing assigned by CRD

Subject index terms
COVID-19; Humans; Residence Characteristics; Viruses; severe acute respiratory syndrome coronavirus 2

Date of registration in PROSPERO
14 October 2020

Date of first submission
13 October 2020

Stage of review at time of this submission

Stage	Started	Completed
Preliminary searches	Yes	No
Piloting of the study selection process	Yes	No
Formal screening of search results against eligibility criteria	No	No
Data extraction	No	No
Risk of bias (quality) assessment	No	No
Data analysis	No	No

The record owner confirms that the information they have supplied for this submission is accurate and complete and they understand that deliberate provision of inaccurate information or omission of data may be construed as scientific misconduct.
The record owner confirms that they will update the status of the review when it is completed and will add publication details in due course.

Versions
14 October 2020